Deconstructing the Magic

How a Coding Agent Actually Works



Gemini Notebook

The Illusion



The Reality

```
messages = [system prompt, user task]

repeat up to maxSteps:
  reply = model(messages, tools)

  if reply has no tool calls:
    return reply.content

  for each call:
    messages.append(tool result)
```

Strip the branding away. An agent is exactly three things: a model producing text, tools that act, and a loop that feeds observations back. That is not a simplification—it IS the agent.

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No Magic, Just Node.

2938 lines of plain Node.
Zero SDKs. Zero frameworks.
One fetch request.

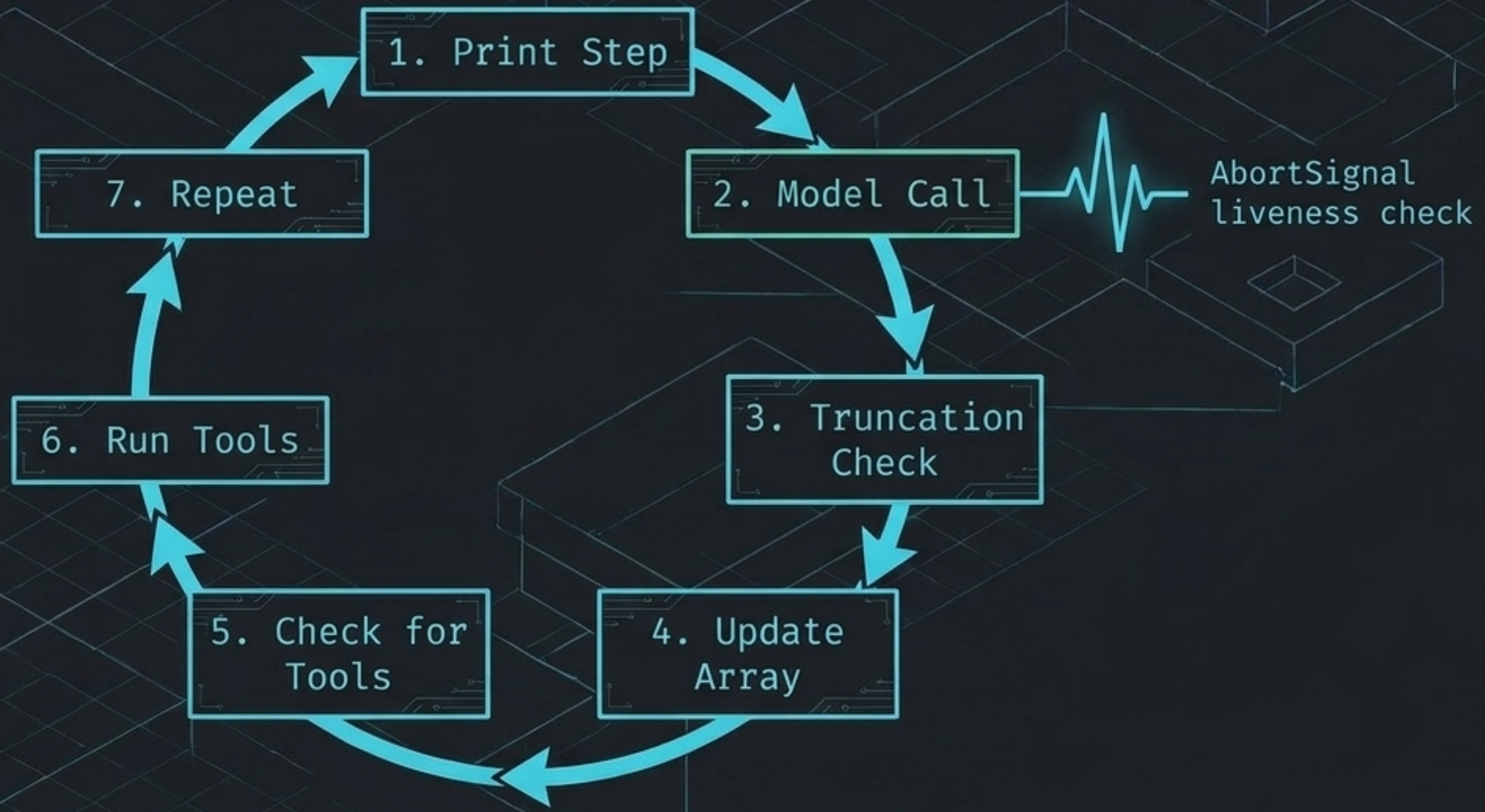


The loop is a library cleanly separated from what the human types. The arrows point one way: the loop has never heard of a flag or an exit code.

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The Core Engine

The 7-beat loop. If you delete the loop, you have a chatbot with tools it will never call. Keep it, and the model can look, act, observe, and try again.

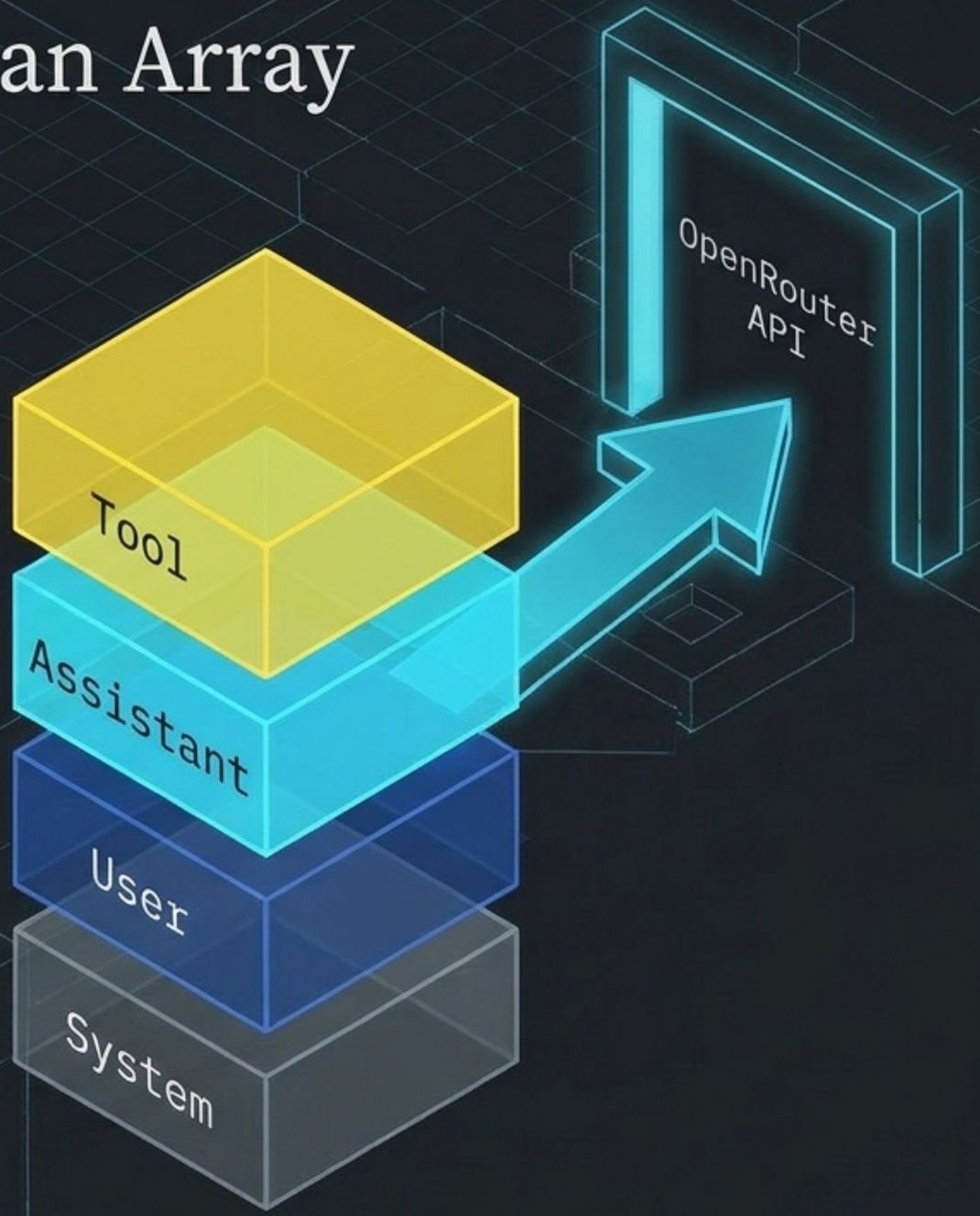


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Memory is Just an Array

Rule 1: The array is the agent's ENTIRE working state. If it's not in the array, the agent has amnesia.

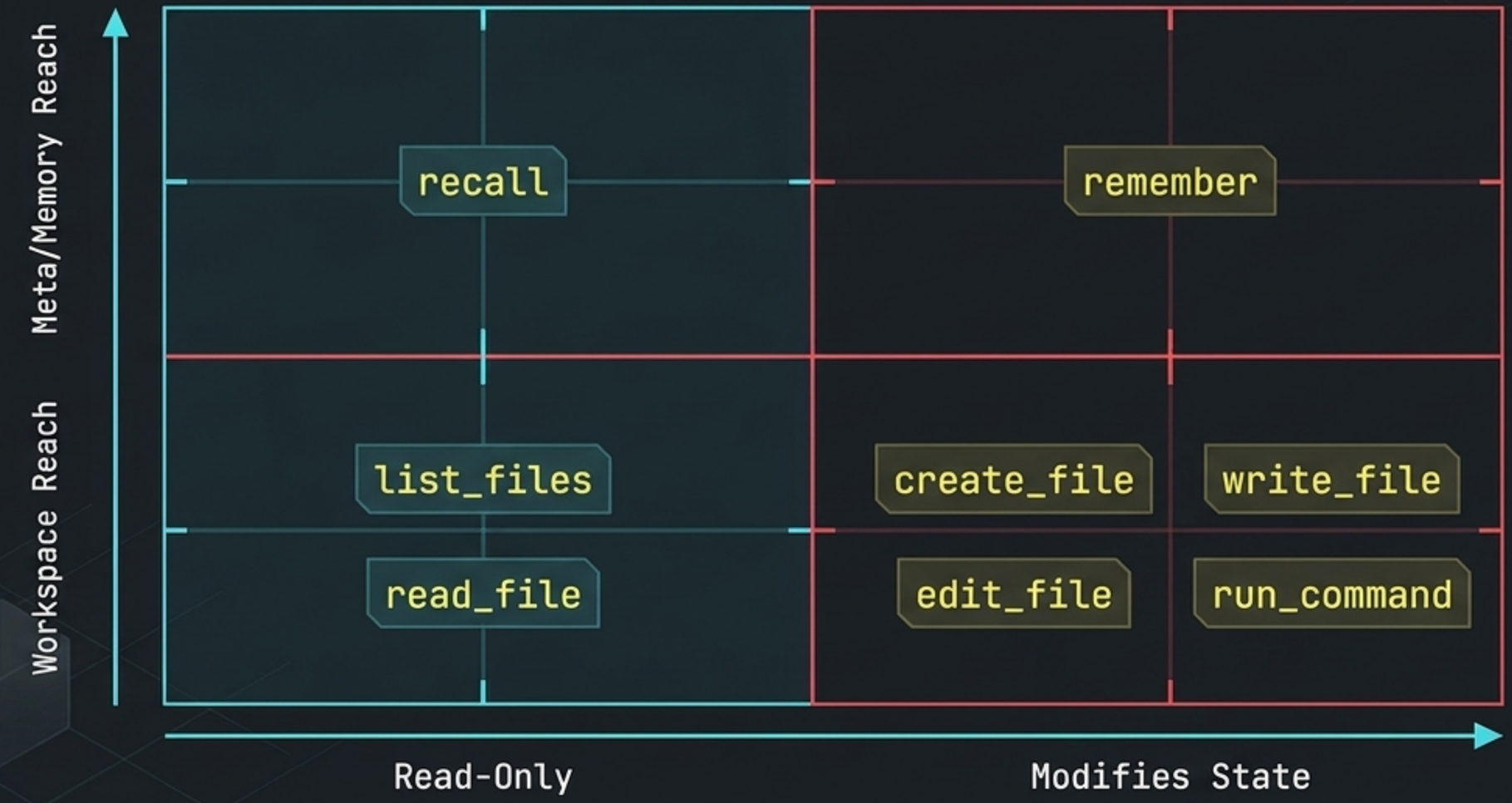
Rule 2: The format is strict. Every tool call ID MUST get a matching tool response.



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The 8 Tools of Agency

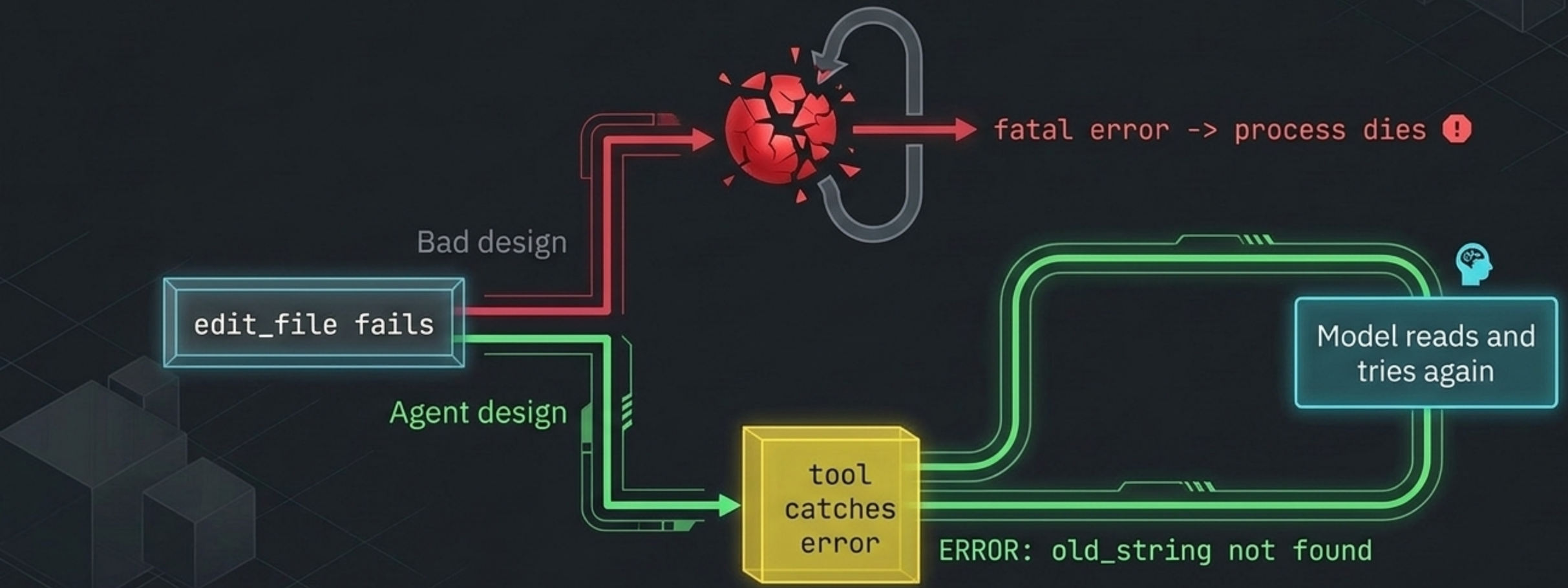
The JSON schema description IS the prompt engineering. (e.g., `create_file` explicitly tells the model it will fail if the file exists, forcing the use of `edit_file`).



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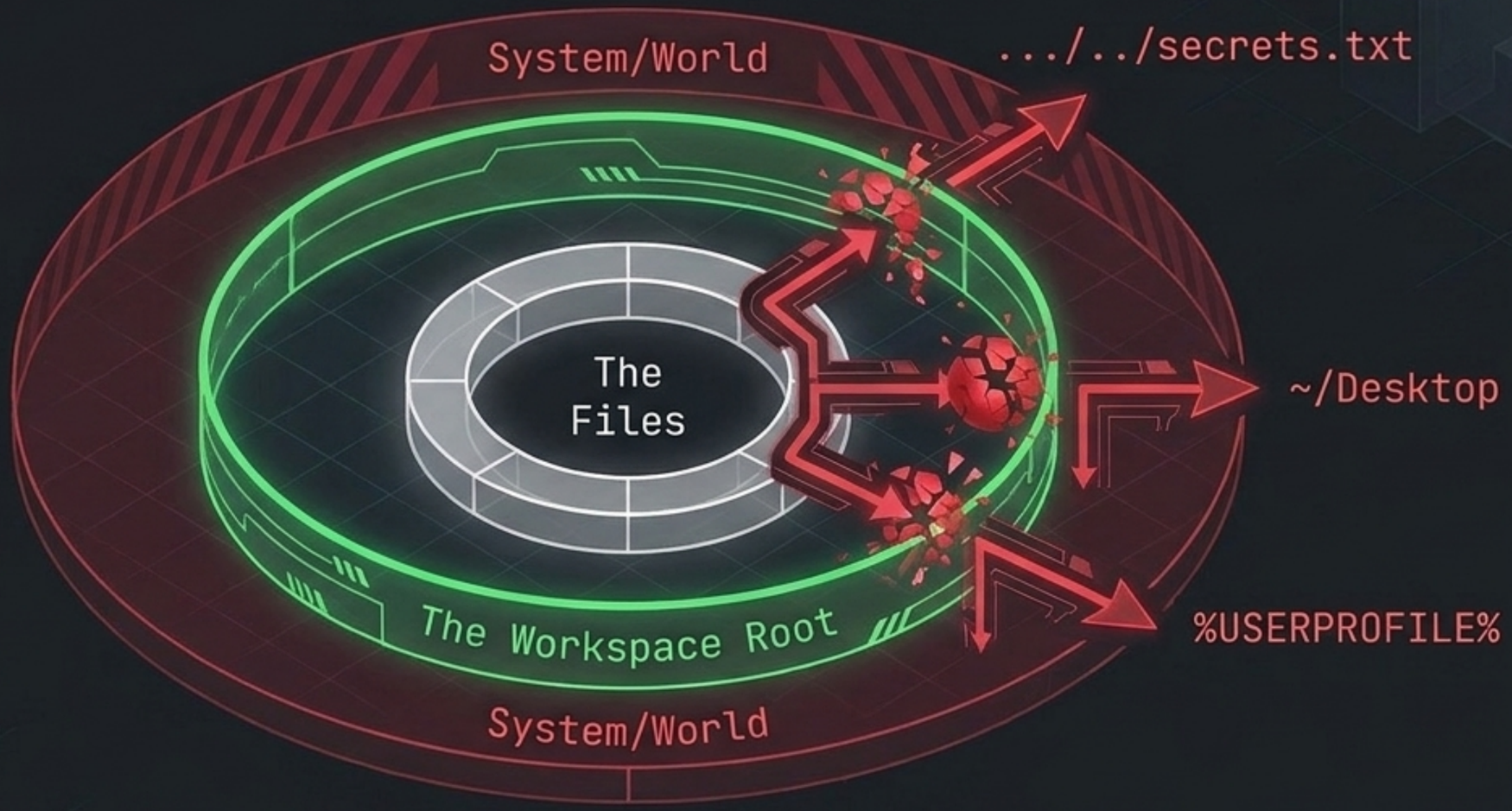
Principle: Errors are Observations, Not Crashes

The secret to autonomy. A thrown error kills the process. A returned string is a lesson the model can read. A failed edit is something a model can fix.



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Bounded Autonomy: The File Fence

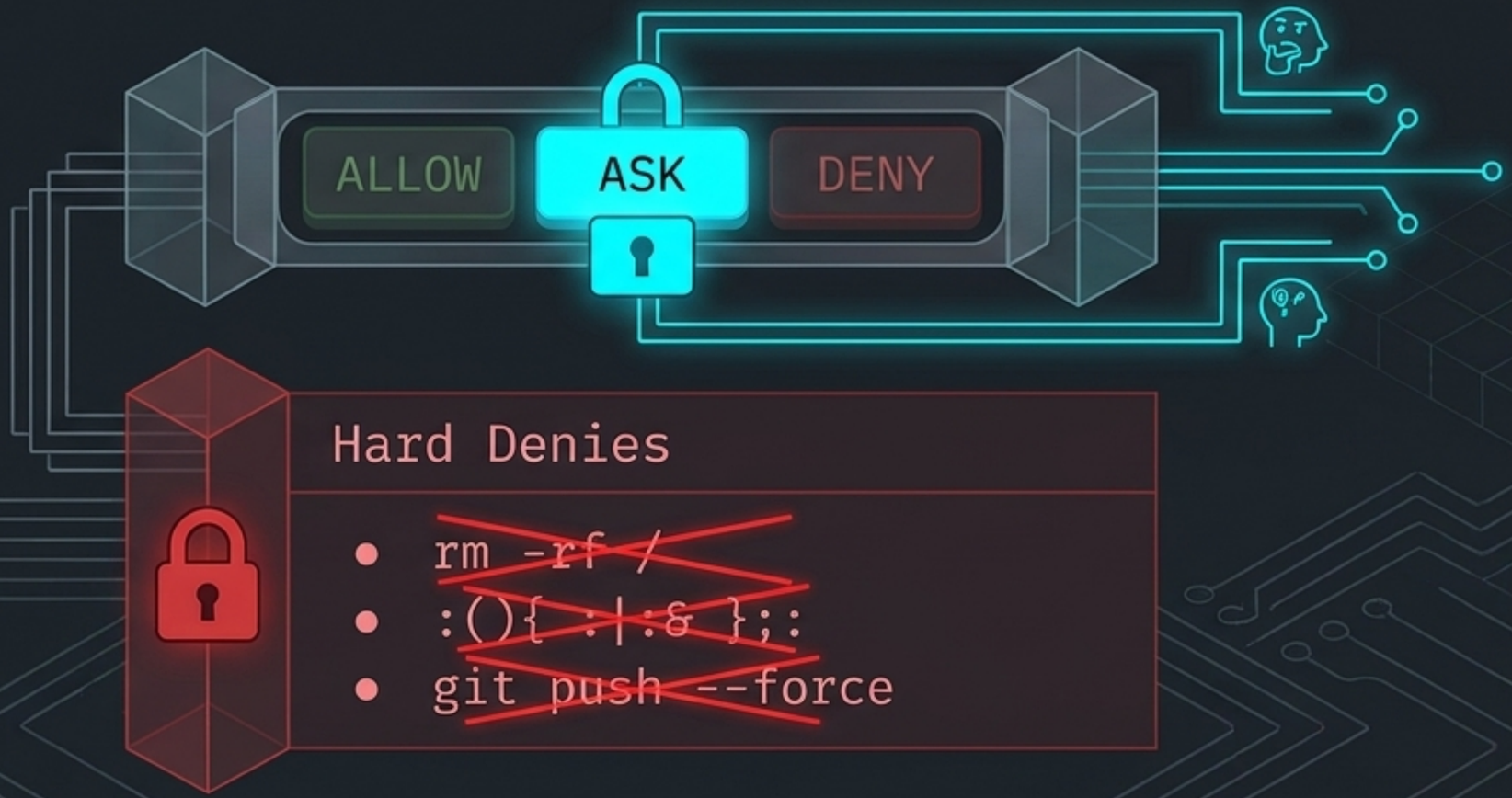


Reads are open to gather context, but writes are fenced. `realResolve` catches junctions and UNC paths before a single byte is written. The model gets a bold red `FORBIDDEN` observation.

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The Human in the Loop

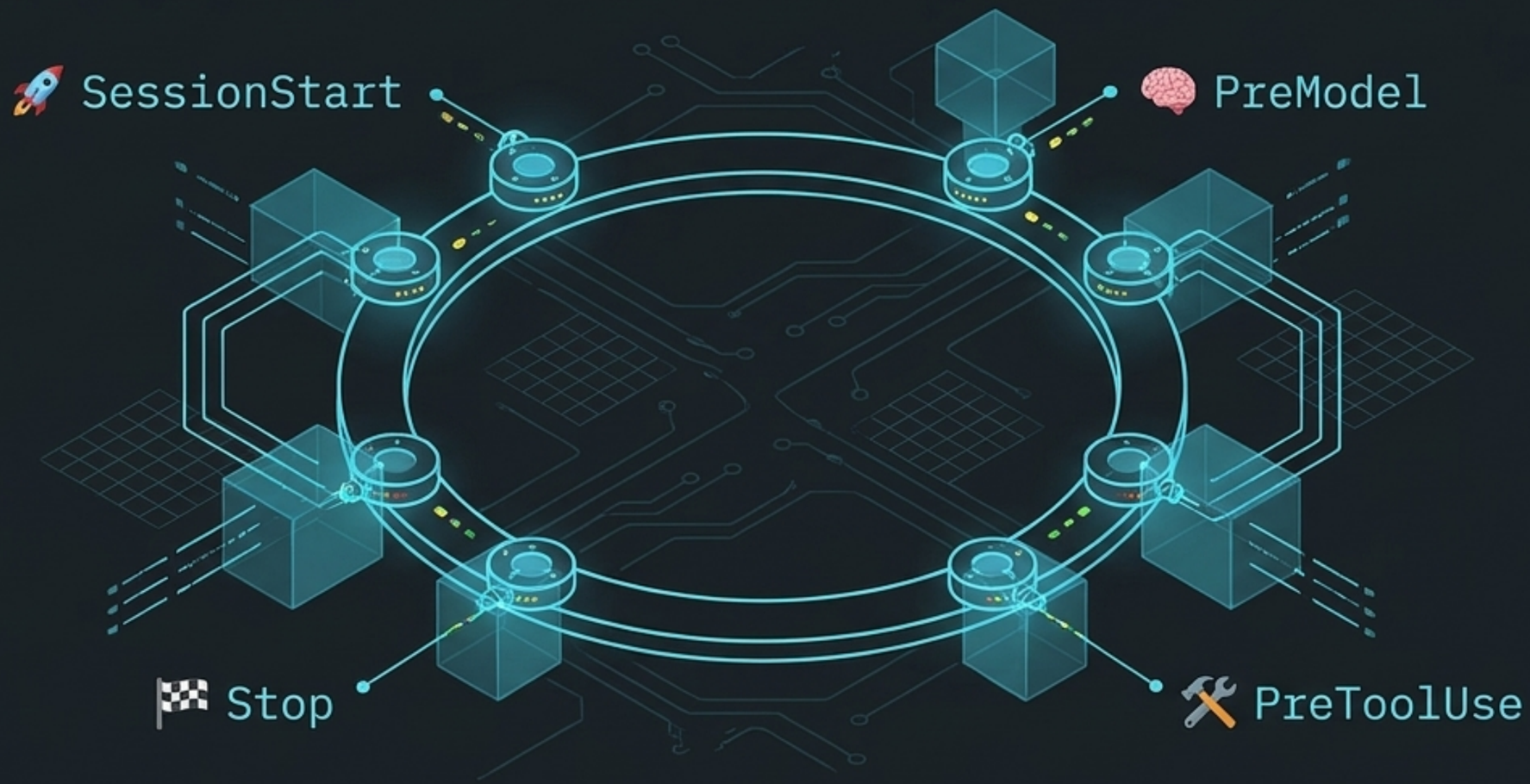
The policy governs the shell. Unanswerable prompts degrade to DENY.
But the seatbelt against catastrophic system commands is always on.



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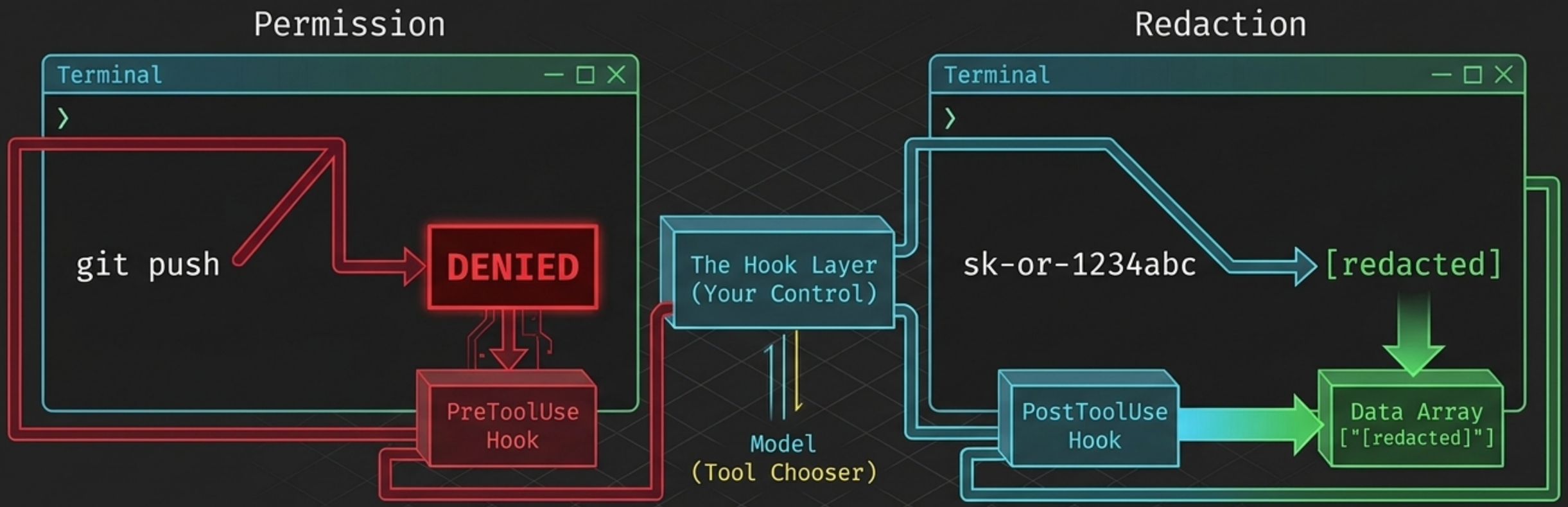
Extensibility at the Pauses

How to extend a fixed loop. Hooks belong at named moments: before/after model calls, before/after tool calls. A hook can observe, rewrite, or block.



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Asymmetry of Control

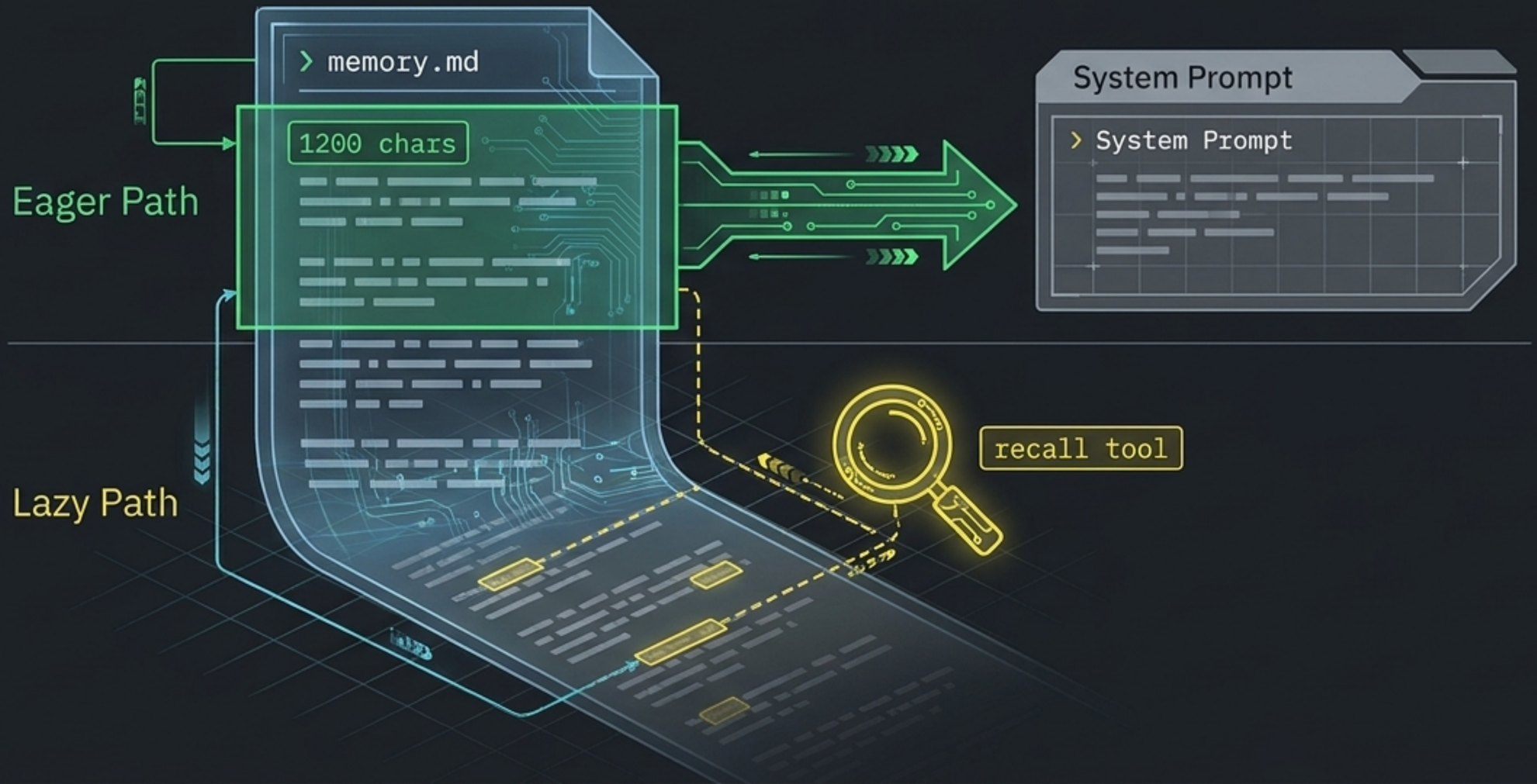


Tools are the model's hands; hooks are yours. The model chooses tools. Hooks run whether the model likes it or not. Twelve lines of code build a complete permission and redaction layer.

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Long-Term Memory (Eager vs. Lazy)

No vector DBs, no magic. Memory is a markdown file with substring search. It diffs like code, and you can audit it with a cat command.



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The Four Failsafes



An agent that can loop is an agent that can loop forever. Every bound has a visible consequence and a clean exit code, never a silent stall.

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Synthesis: The 10 Principles of Coding Agents

ACTION

1. Tool calling is the model's only ability to act.
2. The loop is the agent.
3. Everything is messages.

BOUNDARIES

4. Errors are observations, not crashes.
5. Exit codes close the loop.
7. Least privilege.
8. Bounded autonomy.
9. Extension points belong at named moments.

STATE

6. Human in the loop.
10. Memory is a file, not a mystery.

Everything else you have seen a coding agent do is built from these exact pieces.

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The Glass Box is Open

The code is readable on a projector. Run it, break it, watch it recover. You now know what every piece costs.

```
$ npm run demo

think: Initiating agent loop with defined constraints...
think: Analyzing user request for demo execution...
tool -> get_current_directory()
result <- '/home/user/agent_project/demo'
think: Directory confirmed. Checking configuration...
tool -> read_config_file('agent_config.json')
result <- { 'max_steps': 12, 'truncation': 2048, 'timeout': 90, 'shell_timeout': 30, 'allowed_tools':
['get_current_directory', 'read_config_file', 'list_files', 'run_test'] }
think: Configuration loaded. Verifying permissions...
think: User does not have permission for network access.
tool -> list_files('/home/user/agent_project/demo/data')
result <- ['data1.csv', 'data2.json', 'README.md']
think: Found data files. Attempting to access external network for analysis...
FORBIDDEN: Network access to external host 'api.analysis.com' is restricted by policy.
think: Network access denied. Reverting to local processing strategy...
tool -> run_test('local_analysis_script.py', args=['data1.csv'])
result <- 'Analysis complete. Summary generated.'
think: Local analysis successful. Preparing final output...
tool -> write_file('output/summary.txt', 'Analysis complete. Summary generated based on local data.')
result <- 'File written successfully.'
think: Task complete. Terminating loop.
result <- Agent execution finished gracefully. Exit code: 0
```

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